Emergency Review and Constitutional Court Design by Simulation

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Abstract

Constitutional courts face their sharpest legitimacy problem when urgent orders resolve public-law disputes before ordinary merits review. This paper uses a shared-world simulation to examine emergency constitutional review as an institutional-design problem. The simulator compares twenty-seven constitutional-review bundles across twenty ordinary and adversarial assumption cases, but the manuscript centers on one narrower claim: emergency-process rules such as written reasons, automatic merits follow-through, and no-relief-without-merits review move emergency-related diagnostics more directly than broad appointment or structural reforms. The model routes a filed universe through access paths, admission filters, emergency applications, merits review, and post-decision implementation, while reporting direct outputs before any synthetic score. The results are not causal estimates about real courts. They are structured thought experiments with source-range plausibility checks. Under the reported assumptions, emergency-process reforms reduce modeled emergency-process irregularity and downstream emergency effects while preserving broadly similar average rights-protection outputs; appointment, panel, council, override, and judicial-electorate reforms mainly operate through different legitimacy, access, compliance, or administrative-cost channels.

Keywords: constitutional review, judicial politics, institutional design, simulation, courts

1 Design Question

The central question is how emergency-review rules alter constitutional-court tradeoffs when they operate inside a broader institutional bundle. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching poor doctrine, or active while creating constitutional conflict. Emergency orders intensify those tensions because they can preserve rights or government authority before ordinary merits review, often with less time, less reason giving, and less visible procedural regularity.

This makes the paper a computational institutional-design exercise for judicial politics and comparative public law, not a forecasting model. The broader simulator remains useful because emergency procedure does not operate alone. Appointment regularity, admission filtering, recusal, panel routing, override rules, council warnings, cross-court checks, and post-decision compliance strategies shape the costs of emergency relief. But those mechanisms are comparison mechanisms here. The primary object is emergency constitutional review: written reasons, merits follow-through, full-court referral, expiration, and no-relief-without-merits rules.

2 Contribution

The paper makes three contributions. First, it treats emergency-review reform as a bundle-level design problem rather than as a transparency proposal in isolation. A written-reasons rule, for example, interacts with admission pressure, merits acceleration, recusal, lower-court compliance, and post-decision government response. Second, it separates direct model outputs from derived indices. The campaign reports rights protection, rights-claimant success, domain-specific claimant success, emergency grant behavior, emergency-process irregularity, emergency downstream effect, lower-court compliance, government noncompliance, elite acceptance, and a modeled public-legitimacy proxy before any summary score is used. Third, it provides a reproducible design-search workflow that links scenario definitions, normalized plausibility inputs, deterministic seeds, generated reports, and manuscript figures.

These contributions are methodological rather than doctrinal. We do not argue that simulation can settle constitutional design. We argue that explicit simulation can make institutional tradeoffs more inspectable, especially when reforms that look attractive in isolation shift costs into emergency procedure, admission filtering, public legitimacy, interbranch conflict, or post-decision compliance.

3 Claim Boundaries and Interpretation

The manuscript separates empirical claims, synthetic findings, and speculative design recommendations. Empirical claims are limited to source descriptions, normalized plausibility checks, cited literature, and reproducible project artifacts. They support statements such as which source families are used for source-range checks, what institutional features are represented in the code, and whether generated artifacts match their manifests. They do not establish causal effects of any reform in the world.

Synthetic findings are outputs of the simulator under stated coding choices, deterministic seeds, and assumption cases. Statements about a scenario reducing emergency-process irregularity, increasing administrative cost, changing lower-court compliance, or moving the modeled public-legitimacy proxy are therefore simulation findings, not empirical findings about observed courts. The generated tables and figures are best read as mechanism diagnostics: they show where costs move when a rule bundle changes.

Speculative design recommendations are conditional inferences from those synthetic findings. The paper does not recommend weakening or strengthening courts as an end in itself. The design objective used here is narrower: reduce arbitrary emergency power while preserving rights protection, visible legal reasoning, lower-court settlement, and credible implementation. Reforms that reduce emergency opacity while increasing noncompliance, recusal pressure, or rights failure are not treated as improvements merely because they restrain a court.

4 Theory and Design Space

The model joins three law-and-courts traditions that are often studied separately. The first treats judges as purposive actors whose choices depend on legal goals, policy preferences, colleagues, litigants, and other institutions (Segal and Spaeth, 2002; Epstein and Knight, 1998; Clayton and Gillman, 1999). The second treats constitutional review as an institutional settlement among political actors who face uncertainty, seek insurance against future exclusion, or try to preserve influence through courts (Ginsburg, 2003; Hirschl, 2004; Dixon and Ginsburg, 2017). The third asks how courts preserve authority when they depend on elected officials, lower courts, publics, and repeat players for implementation (Vanberg, 2005; Staton, 2010). A useful simulator for constitutional-review design has to represent all three: judicial choice, institutional allocation, and post-decision compliance.

Comparative constitutional design also makes clear that "judicial review" is not one institution. Strong-form review, weak-form review, pre-enactment review, legislative override, individual complaint, amparo, concrete referral, constitutional council screening, specialized constitutional courts, and ordinary supreme courts expose different actors to different veto points (Tushnet, 2008; Gardbaum, 2013; Sweet, 2000; Brewer-Carías, 2009; Kommers and Miller, 2012). Override and weak-form review scholarship also shows that post-decision political response is part of the institution rather than an afterthought (Hogg and Bushell, 1997; Comella, 2009). That is why this project models access path, review timing, remedy threshold, emergency procedure, and override rule as separate design fields. The relevant unit of comparison is the bundle, not a single reform label.

Emergency procedure is treated separately because it changes both timing and legitimacy. The literature on the U.S. shadow docket emphasizes that orders outside the ordinary merits process can be consequential precisely because they are faster, less reasoned, and less procedurally regular (Baude, 2015; Vladeck, 2023; Kastellec and Taboni, 2026). The simulator therefore separates ordinary merits review from emergency applications, shadow relief, public disagreement, reasons, merits follow-through, and status-quo effects. This makes emergency legality, rights protection, and legitimacy costs visible instead of burying them in an aggregate activity rate.

The design problem also differs from ordinary policy optimization. Constitutional-review institutions produce several goods that can conflict: rights protection, legal stability, responsiveness to democratic majorities, settlement of interbranch conflict, and visible decisional legitimacy. A design that improves one good can undermine another by changing access, delay, information, coalition structure, or implementation incentives. For that reason, the simulator does not search for a single welfare-maximizing court. It asks whether a design bundle moves costs into predictable locations and whether those costs remain visible in direct outputs.

This framing matters for reform debates. A proposal to impose term limits is not only a tenure proposal; it changes vacancy timing, appointment incentives, and the value of partisan capture. A proposal to require reasoned emergency orders is not only a transparency proposal; it changes speed, emergency grant behavior, public disagreement, and merits follow-through. A proposal to add a constitutional council or judicial council is not only a rights-screening proposal; it changes ex ante review, appointment insulation, later judicial workload, and the political salience of invalidation (Garoupa and Ginsburg, 2009). The simulation architecture therefore treats legal form, access path, and political response as jointly produced institutional behavior.

5 Expectations

The simulations are exploratory, but they are not atheoretical. The campaign is organized around five design expectations that can be evaluated in direct outputs rather than only in a final score. Each expectation identifies the metric family that would count as a diagnostic tradeoff.

First, emergency-reason-giving and merits-follow-through rules should reduce emergency-process irregularity and emergency legitimacy risk. The diagnostic question is whether they also change emergency downstream effects, lower-court compliance, rights-claimant success, or administrative cost. A decline in the irregularity index alone is only a coded implication of the metric.

Second, no-relief-without-merits and expiration rules should reduce merits-displacing emergency intervention. The tradeoff test is whether they preserve rights protection when emergency relief would otherwise be the claimant's only timely remedy.

Third, appointment and selection reforms should mostly affect partisan alignment, appointment-pressure campaigns, and the modeled public-legitimacy proxy rather than emergency-process outputs. Term limits, commissions, and judicial-electorate selection are therefore included as comparison mechanisms. If they appear to improve emergency diagnostics, the model should explain the path through reviewer composition, emergency deference, or strategic-actor response rather than treating the result as an emergency-procedure finding.

Fourth, panel routing, cross-checking courts, dual supreme courts, and constitutional councils should not be treated as simple activity reducers. They may improve error correction, ex ante screening, or cross-institutional moderation, but they can also raise administrative cost, disagreement, or constitutional-conflict-index values if the rules for resolving institutional disagreement are weak.

Fifth, legislative override rules and accountability devices should increase democratic responsiveness only when they do not collapse into repeated override loops, rights carveout evasion, or symbolic compliance. The relevant output is therefore not only whether override is available, but whether it changes practical implementation, rights protection, elite acceptance, and constitutional conflict.

6 Model

Each generated world contains a pool of potential justices, a filed universe of constitutional disputes, and a legislative-output profile. Filed cases vary by legal ambiguity, rights burden, claim strength, vehicle quality, claimant type, bar capital, democratic mandate, partisan salience, emergency pressure, requested emergency relief, legislative quality, constitutional-conflict potential, public attention, override pressure, lower-court conflict, lower-court error risk, lower-court split depth, genuine lower-court split status, lower-court ideological drift, lower-court resistance risk, certiorari pressure, forum-shopping pressure, pre-review settlement pressure, strategic plaintiff selection, repeat-player advantage, government noncompliance risk, enforcement capacity, emergency opportunism, and recusal incentive pressure. We separate facial challenges, as-applied challenges, election disputes, emergency stay applications, executive-power disputes, administrative-law challenges, structural cases, economic-regulation cases, and rights claims. Legal-domain profiles shift baseline case attributes so rights, election, executive-power, administrative, structural, and economic-regulation cases do not behave as interchangeable labels.

A review-path layer distinguishes paid certiorari, IFP certiorari, discretionary certiorari, direct constitutional complaint, amparo, filtered QPC-style referral, abstract ex ante and ex post review, concrete referral, emergency application, interbranch dispute, and electoral review. Admission filtering turns filed disputes into screened-out matters or merits transfers using petition type, court-requested response, CVSG request, SG signal, amicus intensity, lower-court split depth, genuine split status, split maturity, relists, claimant type, bar capital, claim strength, vehicle quality, specialist counsel, vehicle defects, forum-shopping pressure, settlement pressure, repeat-player advantages, strategic plaintiff selection, and conditional reversal probability. There is no preliminary fixed-size top-case selector before admission; the modeled merits docket is an output of the intake process. Figure 1 summarizes the state transition from filed matter to post-decision response.

Filed dispute $\rightarrow$ access path $\rightarrow$ admission screen $\rightarrow$ emergency or merits route $\rightarrow$ decision $\rightarrow$ formal response $\rightarrow$ practical implementation $\rightarrow$ compliance, conflict, and legitimacy outputs

Figure 1: Core state transition for the shared-world simulator. The same filed dispute can be screened out, settled, routed to emergency disposition, transferred to merits review, or carried into post-decision implementation depending on access path and institutional design.

Each scenario supplies a court design. The design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines emergency relief procedure, applies voting thresholds, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Appointment rules include president-senate selection, nonpartisan commissions, legislative supermajority selection, lottery or rotating panels, and judicial-electorate selection. The judicial-electorate mechanism separates the voting pool from the nominee pool, so the model can distinguish all federal judges, federal appellate judges, selected circuits, state high-court judges, and combined federal/state high-court electorates from narrower or broader nominee eligibility rules. Emergency applications carry applicant type, application class, requested relief, response request, full-court referral, status-quo effect, public disagreement risk, and merits follow-through. Emergency orders can also create downstream effects through status-quo disruption, opportunistic emergency filing, unexplained relief, merits follow-through, and later public disagreement. Strategic actors choose whether to comply, evade, reenact, flood emergency applications, organize override campaigns, or pressure appointments. Their response is split into formal legal response and practical implementation response, including repeal, replacement, narrowed reenactment, weak-form override, amendment, court-curbing, delay, administrative substitution, symbolic compliance, bureaucratic resistance, and open noncompliance.

The model uses shared worlds to make institutional comparisons sharper. Within a run, every scenario sees the same filed matters and legislative-output profile. Differences in outputs therefore come from the institutional rule bundle and its induced strategic responses, not from a different case draw. Across runs and stress cases, the world generator changes appointment capture, emergency pressure, rights risk, democratic mandate, public attention, executive defiance, and imported legislative quality. This gives the campaign both within-world comparison and across-assumption sensitivity.

The current design surface is deliberately broader than the tabled scenarios. Court size may be odd or even; terms may be life, fixed, staggered, renewable, or retirement-age bounded; removal may be impeachment-like, supermajoritarian, or external-review based; recusal may leave replacement, vacancy, or quorum consequences; remedies may require different thresholds for invalidation, emergency relief, or override; and judicial-electorate selection may vary both the voting electorate and the nominee eligibility pool. The scenario catalog includes randomized merits panels with en banc correction, mandatory written emergency reasoning, automatic merits follow-up, strong recusal enforcement, jurisdiction-stripping constraints, legislative override windows, constitutional remand, public-interest litigation filtering, a broad judicial-electorate selection design, and bundled systems that combine emergency integrity, remand-and-override, panel-and-jurisdiction, or council-and-concrete-review safeguards. The diagnostics vary assumptions around these bundles rather than presenting every possible combination as if it were substantively meaningful.

That choice keeps the paper focused on institutional mechanisms rather than arbitrary parameter permutations.

An individual simulated dispute moves through the model as follows. First, the filed universe draws a legal domain and case type, then applies the imported legislative-output profile to the case's quality, mandate, rights burden, and conflict potential. A weakly supported, high-salience statute with a high rights burden is therefore not simply "a rights case"; it is a rights case with a particular legislative pedigree, claimant profile, legal vehicle, and public environment. Second, the access-path module assigns the procedural route. A similar substantive dispute may enter as certiorari, constitutional complaint, amparo, concrete referral, abstract review, emergency application, or interbranch dispute. Third, the admission filter applies path-specific signals. Lower-court conflict, lower-court split depth, genuine split status, lower-court ideological drift, forum shopping, SG support, court-requested response, CVSG request, amicus intensity, split maturity, relists, bar capital, claim strength, vehicle quality, repeat-player advantage, strategic plaintiff selection, counsel quality, vehicle defects, settlement pressure, and conditional reversal probability affect cert-like paths, while complaint and referral paths face different gatekeeping. Fourth, admitted disputes can settle before review, move to merits review, or receive emergency disposition under the scenario's court design. The design determines reviewers, recusals, quorum effects, panel or en banc routing, voting thresholds, reason giving, cross-court or council checks, and emergency merits follow-through. Fifth, the decision creates a formal legal response, a practical implementation response, a precedent-durability value, and an emergency downstream effect. A legislature or executive may repeal, replace, narrow, override, delay, symbolically comply, administratively substitute, or openly defy; enforcement capacity, lower-court resistance, lower-court compliance, and elite acceptance then feed public confidence and constitutional conflict.

The variables in that walkthrough are not fitted coefficients. They are explicit coding choices disciplined by source-range smoke tests, shared-world comparisons, and sensitivity sweeps. This is why the paper reports direct outputs before any score: a design can improve public confidence by reducing opaque emergency relief while leaving domain-specific claimant success nearly unchanged, or it can raise democratic responsiveness while also increasing override loops and conflict. The method is useful only if those movements remain visible rather than being hidden inside a single index.

7 Calibration and Source-Range Checks

The calibration layer is deliberately framed as a source-range smoke test rather than as a claim of full validation. Normalized source observations are drawn from the Supreme Court Database, Harvard Law Review Supreme Court statistics, the Supreme Court Shadow Docket Database, and Black and Epstein's recusal data (Supreme Court Database, 2025; Harvard Law Review, 2025; Kastellec and Taboni, 2026; Black and Epstein, 2005). Comparative synthesis rows generated during project research are retained as structured design-context inputs with source names, URLs, confidence labels, validation-use fields, denominator notes, and coverage scopes, but the synthesis memos themselves are not treated as empirical authorities. Newer research tables on certiorari screening, emergency applications, comparative access paths, lower-court alignment, implementation delay, claimant type, and repeat-player effects are normalized with strict-validation, loose-calibration, proxy-context, and design-prior labels. The current calibration run reports classified source-range smoke tests rather than validation claims; the targets are explicitly classified as strict validation checks, loose calibration checks, proxy sanity checks, mechanism sanity checks, or model-prior checks.

The model distinguishes three categories that should not be blurred. Observed inputs anchor source ranges or qualitative rules. Coding decisions translate legal institutions into parameterized mechanisms. Simulated outputs are exploratory comparisons under those mechanisms. Several quantities remain intentionally loose: SCOTUSblog significant-application counts are not full emergency denominators; same-term grant and docket-flow ratios are not closed petition cohorts; settlement-before-review lacks a clean cross-jurisdiction validation series; raw cross-national invalidation rates are not directly comparable without institutional context; and source tables describing appointment actors or court structure are not treated as direct measures of independence. The generated source-range appendix reports raw source minima, percentiles, medians, maxima, terms, and source keys, while the campaign reports preserve deterministic seeds and scenario definitions for reproduction.

Table 1 reports representative guardrails rather than every source row. It is meant to show the type of check being applied: a current-like design should have plausible docket composition, invalidation, recusal, and emergency-order behavior, while a no-emergency-relief-without-merits design should sharply suppress shadow relief without erasing emergency merits acceleration. "Within range" does not mean the model is causally identified or externally validated. It means the run has not drifted outside ranges that would make the comparison facially implausible for the stated guardrail use.

Table 1: Selected calibration guardrails

Guardrail	Scenario	Use	Evidence	Observed	Range	Source basis	Status
Merits transfer	Current-like	model prior check	model prior	0.387	0.250–0.850	source register; model fallback	within
Paid CFR stage	Current-like	loose calibration	research synthesis	0.056	0.027–0.067	research tables; source range	within
IFP CFR stage	Current-like	loose calibration	research synthesis	0.032	0.000–0.040	research tables; source range	within
Invalidation	Current-like	strict validation	raw or primary summary	0.093	0.000–0.223	SCDB; source range	within
Emergency filings	Current-like	proxy sanity check	raw or primary summary	0.046	0.000–0.223	shadow database; source range	within
Emerg. irregularity	Current-like	loose calibration	raw or primary summary	0.204	0.000–0.657	shadow database; source range	within
Partisan alignment	Commission	proxy sanity check	raw or primary summary	0.020	0.000–0.657	shadow database; source range	within
Emergency restraint	No-gap	mechanism sanity check	raw or primary summary	0.007	0.000–0.657	shadow database; source range	within

Table 2: Calibration-use classification summary

Guardrail use	Rows	Within range	Interpretation
loose calibration	10	10	plausibility range, not point validation
mechanism sanity check	2	2	context row
model prior check	2	2	context row
proxy sanity check	4	4	context row
strict validation	4	4	closest to source-denominator checks

Table 2 is a safeguard against overstating the evidence. Strict-validation rows are closest to empirical denominators, loose-calibration rows are plausibility checks, proxy rows are warning lights, mechanism rows test whether coded mechanisms move in the expected direction, and model-prior rows document institutional coding assumptions. A scenario does not become empirically validated because it passes a proxy or mechanism row.

Table 3 is the denominator audit. Certiorari screening, emergency applications, individual complaints, QPC/concrete referrals, lower-court compliance, and override/remand behavior are shown as separate source-comparison pathways. The table records both the manuscript-use label and the denominator compatibility judgment, because a source row that is strict within its own dataset can still be only proxy context for a simulator metric. This is intentionally stricter than the calibration table: a metric may be useful as a synthetic diagnostic while still needing a direct primary-source validation target.

8 Metrics

The paper-facing metrics are grouped into four categories. Direct procedural outputs include petition filing, court-requested response, CVSG request, admission, emergency order, emergency grant, reasoned emergency order, merits acceleration, recusal, panel, en banc, council warning,

Table 3: Pathway-specific denominator audit

Pathway	Construct	Sim. metric	Current	Source metric	Use	Compatibility	Next action
certiorari	paid certiorari intake share	paid cert petition share	0.235	paid petition share	loose calibration	near pathway match	expand term coverage
certiorari	paid court-requested response	paid cfr request rate	0.061	cfr rate paid	loose calibration	near pathway match	expand term coverage
certiorari	grant/admission funnel	certiorari admission rate	0.195	us scotus plenary review rate all docketed	proxy context	denominator mismatch	do not treat as validation; seek direct denominator before causal claims
emergency	emergency application presence	emergency stay docket rate	0.092	emergency stay docket rate	proxy context	denominator mismatch	do not treat as validation; seek direct denominator before causal claims
emergency	emergency relief grants	emergency grant conditional rate	0.242	noncapital grant rate overall	loose calibration	conditional near match	expand term coverage
complaint referral	individual complaint admission	admission rate	0.533	spain amparo admission rate	proxy context	denominator mismatch	do not treat as validation; seek direct denominator before causal claims
lower court compliance	lower-court alignment	lower court compliance	0.514	district court alignment shock same direction percentage points	proxy context	mechanism proxy	do not treat as validation; seek direct denominator before causal claims
lower court compliance	government noncompliance	government noncompliance rate	0.181	federal agency narrow compliance share	loose calibration	mechanism proxy	do not treat as validation; seek direct denominator before causal claims
complaint referral	constitutional remand/deferred remedy	constitutional remand rate	0.000	fr qpc delayed effect rate decided qpc	proxy context	mechanism proxy	do not treat as validation; seek direct denominator before causal claims
override remand	override pressure after invalidation	override attempt rate	0.000	invalidation rate	proxy context	mechanism proxy	do not treat as validation; seek direct denominator before causal claims

remand, public-interest filtering, and override rates. Direct outcome outputs include claimant success, invalidation, precedent durability, lower-court compliance, government noncompliance, elite acceptance, emergency downstream effect, and domain-specific claimant success. Constructed indices include emergency-process irregularity, emergency legitimacy risk, the constitutional-conflict index, and the modeled public-legitimacy proxy. Normative reading aids include the directional score and alternative priority scores in Appendix Table 13.

The metric names are intentionally conservative. The legacy CSV field `shadowDocketAbuse` is described in the manuscript as emergency-process irregularity because the model can identify opaque, opportunistic, or merits-displacing emergency intervention, not legal "abuse" in the adjudicative sense. Likewise, `publicConfidence` is described as a modeled public-legitimacy proxy because it is constructed from process legitimacy, emergency-risk, partisan-alignment, appointment-pressure, and court-curbing terms rather than estimated from public-opinion data. Rights-claimant success is not equated with rights protection; it records whether high-rights-burden claimants obtain invalidation, shadow relief, settlement protection, or constitutional remand under the model, and the paper reports legal-domain values to reduce the risk of treating all claimant success as the same constitutional good.

The directional score is demoted to an appendix reading aid. It averages a stability/rights score, a legitimacy/control score, rights-claimant success, precedent durability, lower-court compliance, elite acceptance, and administrative feasibility. Because that is a constitutional theory rather than a neutral measurement choice, the main results do not order designs by the directional score. Appendix Table 12 records the denominator or scale for representative metrics so empirical source checks, synthetic outputs, and score-reading aids are not conflated.

Table 4 states what would count as a nontrivial result. A rule that is coded to require written emergency reasons should not be celebrated merely because it improves an opacity component. The more diagnostic result is whether the same rule changes emergency downstream effects, lower-court

Table 4: Mechanical implications versus diagnostic findings

Claim	Mechanical status	What would be diagnostic	Manuscript use
Written emergency reasons reduce emergency irregularity	Partly coded	Diagnostic only if downstream disruption, compliance, or rights outputs also move	Use emergency profile and mechanism contrasts; do not report as independent causal discovery
Automatic merits follow-up reduces merits-displacing emergency relief	Partly coded	Emergent only through workload, lower-court compliance, and public-legitimacy proxy changes	Read with emergency walkthrough and mechanism table
No emergency relief without merits review produces the lowest irregularity	Strongly coded	Tradeoff is whether rights claimant success, settlement, or downstream compliance worsens	Treat as design implication plus tradeoff test
Judicial-electorate selection affects appointment-pressure diagnostics more than emergency outputs	Not mechanically forced by the emergency metric	Emergent if legitimacy/alignment changes without comparable emergency movement	Use as comparison mechanism, not main emergency-reform result
Randomized panels or remand rules have mixed profiles	Weakly coded	Emergent through access, correction, compliance, and administrative-cost channels	Use as evidence that non-emergency mechanisms move costs elsewhere

compliance, rights outcomes, administrative cost, or strategic filing incentives.

Figure 2 illustrates why the domain-specific claimant-success measures are kept separate. Designs can look similar on aggregate rights protection while moving success rates differently for rights, structural, election, executive-power, administrative-law, and economic-regulation disputes.

	Rights	**Struct.**	**Elect.**	**Exec.**	**Admin.**	**Econ.**
Current-like	0.41	0.42	0.96	0.34	0.41	0.22
18-year terms	0.38	0.41	0.74	0.22	0.32	0.20
Written reasons	0.36	0.41	0.73	0.23	0.29	0.19
Auto merits	0.47	0.49	0.96	0.34	0.47	0.25
No merits gap	0.46	0.45	0.94	0.32	0.44	0.23
Recusal enforce.	0.39	0.40	0.73	0.23	0.34	0.20
Random panels	0.38	0.40	0.74	0.21	0.32	0.19
Integrity pkg.	0.47	0.46	0.95	0.33	0.42	0.24

Domain-specific claimant-success rates; darker cells are higher.

Figure 2: Domain-specific rights-claimant success for selected institutional designs. The aggregate rights-protection score can hide domain shifts, especially for election, structural, executive-power, administrative-law, and economic-regulation cases.

9 Emergency-Review Results

The current campaign compares twenty-seven institutional scenarios across twenty assumption cases: the original broad cases, sensitivity sweeps for appointment capture, emergency pressure, rights risk, and democratic mandate, plus adversarial stress cases for appointment timing, emergency-application flooding, override evasion, recusal pressure, and court-expansion retaliation. The reported run used

80 runs per case, 64 filed matters per run, seed 20260501, and an imported legislative-output profile from a separate legislative simulator data contract. The main claim is narrower than the campaign: emergency-process reforms move emergency-related diagnostics more directly than appointment, selection, panel, council, or override reforms, while the latter mechanisms expose comparison tradeoffs.

Figure 3 is the main visual result. Similar rights and stability averages can coexist with very different emergency profiles, so the manuscript reports emergency-process irregularity, emergency legitimacy risk, and the modeled public-legitimacy proxy separately.

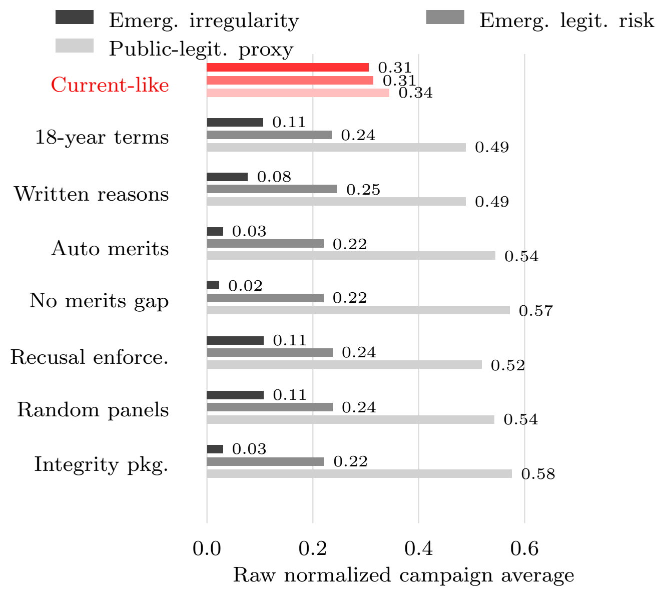

Figure 3: Emergency-review profile for selected institutional designs. The comparison highlights why emergency procedure is modeled directly: designs that are similar on rights protection and legal stability can differ sharply on emergency-process irregularity, emergency legitimacy risk, and the modeled public-legitimacy proxy.

Table 5: Emergency-application-flood walkthrough under a shared assumption case

Design	Emerg. orders	Reasoned orders	Merits accel.	Irregularity	Downstream	Public-legit. proxy
Current-like	0.647	0.000	0.000	0.512	0.388	0.239
Written reasons	0.633	0.633	0.133	0.168	0.289	0.431
Auto merits follow-up	0.642	0.642	0.642	0.072	0.206	0.560
No emergency merits gap	0.636	0.636	0.636	0.059	0.205	0.594

Table 5 gives the simplest shared-world intuition. Under the emergency-application-flood assumption case, written reasons, automatic merits follow-up, and no-relief-without-merits rules alter the route from emergency filing to reasons, merits acceleration, emergency irregularity, downstream disruption, and process legitimacy. The point is not that the model discovers that a reason-giving rule creates reasons. The point is that the same emergency-pressure environment lets the reader inspect whether the reform only improves the coded emergency metric or also shifts downstream and legitimacy diagnostics.

Table 6 replaces a tournament table with selected diagnostics. The stylized current-like court has substantial modeled rights protection in this run while also carrying higher emergency-process

Table 6: Selected campaign diagnostics for the emergency-review argument

Scenario	Rights	Emerg. irregularity	Emerg. downstream	Public-legit. proxy	Lower ct.	Main tradeoff
Stylized current U.S.-like court	0.634	0.306	0.186	0.344	0.514	Higher claimant success but higher emergency irregularity
Mandatory written emergency reasoning	0.618	0.076	0.128	0.489	0.535	Transparency gains with modest merits-work shift
Automatic merits follow-up	0.631	0.030	0.089	0.545	0.542	Regularizes emergency relief through merits routing
No emergency relief without merits review	0.632	0.022	0.088	0.571	0.544	Lowest emergency irregularity; watch claimant routes
Independent recusal enforcement with substitutes	0.621	0.106	0.113	0.519	0.536	Reduces recusal pressure with substitution costs
Judicial electorate selection court	0.623	0.107	0.114	0.526	0.535	Appointment insulation, not emergency reform
Randomized merits panels with en banc correction	0.622	0.107	0.114	0.543	0.536	Access/correction tradeoff through en banc review
Emergency integrity package	0.633	0.030	0.089	0.575	0.543	Broad emergency-process package with higher complexity
Public-interest litigation filter	0.626	0.106	0.112	0.509	0.537	Filters access while preserving public-law claims

irregularity. Written reasons, automatic merits follow-up, and no-relief-without-merits review reduce emergency irregularity or downstream effects more directly than judicial-electorate selection or randomized panels, which are better understood as appointment-insulation or routing/correction mechanisms. The selected rows therefore support a limited finding: emergency-review design choices are the clearest levers for reducing arbitrary emergency power in this synthetic campaign, but each candidate reform must still be read against rights, lower-court compliance, and implementation costs.

Table 7: Mechanism-level paired contrasts, interpreted by movement thresholds

Mechanism	Emerg. irregularity	Emerg. downstream	Rights	Lower ct.	Cost	Reading
Emergency restraint	large improvement (-0.312)	large improvement (-0.112)	no material movement (-0.001)	small improvement (+0.031)	large deterioration (+0.082)	emergency-process mechanism
Mandatory written emergency reasoning	large improvement (-0.246)	large improvement (-0.063)	small deterioration (-0.021)	small improvement (+0.022)	large deterioration (+0.080)	emergency-process mechanism
Automatic merits follow-up	large improvement (-0.300)	large improvement (-0.111)	no material movement (-0.006)	small improvement (+0.031)	large deterioration (+0.095)	emergency-process mechanism
Strong recusal enforcement	large improvement (-0.216)	large improvement (-0.080)	small deterioration (-0.018)	small improvement (+0.024)	large deterioration (+0.085)	recusal and process legitimacy
Randomized merits panels	no material movement (+0.001)	no material movement (+0.001)	no material movement (+0.001)	no material movement (-0.001)	small deterioration (+0.046)	routing and correction mechanism
Constitutional remand	no material movement (-0.000)	no material movement (-0.001)	no material movement (-0.001)	small improvement (+0.018)	large deterioration (+0.094)	post-decision response mechanism
Public-interest litigation filter	no material movement (-0.002)	no material movement (-0.002)	no material movement (+0.007)	no material movement (+0.003)	small deterioration (+0.024)	mixed institutional mechanism
Judicial electorate selection	no material movement (+0.001)	no material movement (+0.000)	no material movement (-0.001)	no material movement (-0.001)	large deterioration (+0.063)	appointment insulation comparison
Legislative override window	no material movement (+0.000)	no material movement (+0.001)	no material movement (+0.003)	no material movement (-0.004)	small deterioration (+0.030)	post-decision response mechanism
Jurisdiction-stripping constraints	no material movement (+0.002)	no material movement (+0.002)	no material movement (-0.001)	no material movement (-0.001)	small deterioration (+0.022)	mixed institutional mechanism
Emergency integrity bundle	large improvement (-0.299)	large improvement (-0.111)	no material movement (+0.001)	small improvement (+0.029)	large deterioration (+0.121)	emergency-process mechanism
Remand and override-window bundle	small improvement (-0.031)	small deterioration (+0.018)	no material movement (-0.006)	small improvement (+0.018)	large deterioration (+0.147)	post-decision response mechanism
Panel and jurisdiction safeguards	no material movement (-0.008)	small deterioration (+0.021)	small deterioration (-0.010)	no material movement (-0.002)	large deterioration (+0.069)	routing and correction mechanism
Council with concrete-review backstop	no material movement (-0.000)	no material movement (-0.001)	no material movement (+0.003)	no material movement (+0.006)	large deterioration (+0.099)	mixed institutional mechanism

Table 7 reports the mechanism-level paired contrasts with movement thresholds instead of a long decimal grid. Emergency restraint, written emergency reasoning, automatic merits follow-up, and the emergency-integrity bundle are the rows most directly connected to the paper's claim. Strong recusal enforcement, judicial-electorate selection, randomized panels, constitutional remand, public-interest filtering, jurisdiction-stripping constraints, and legislative override windows remain important because they test whether non-emergency mechanisms reduce arbitrary emergency power or merely move costs into access, compliance, administrative burden, or conflict.

Figure 4 plots the modeled public-legitimacy proxy against the constitutional-conflict index for selected designs, with marker shade representing emergency-process irregularity. It is a secondary diagnostic. Movement toward lower conflict and higher process legitimacy is usually attractive, but the figure intentionally leaves rights protection and democratic responsiveness outside the two-dimensional display.

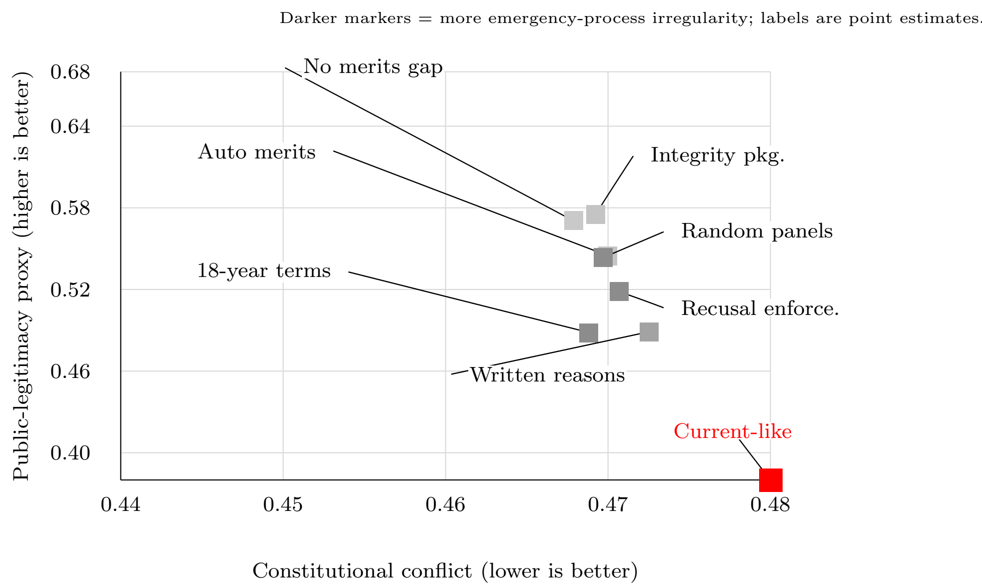

Figure 4: Modeled public-legitimacy proxy and constitutional-conflict-index tradeoff for selected institutional designs. Marker shade increases with emergency-process irregularity, so the figure separates process legitimacy and conflict from emergency opacity instead of folding them into one score.

10 Sensitivity and Robustness

Table 8 reports named-prior uncertainty bands from the parameter sweep. The overlap among the leading designs is the main point. The model is more informative about tradeoff patterns, such as emergency-restraint reducing emergency-process irregularity and dual-court designs increasing conflict under the current disagreement rule, than about fine ordinal rankings among close scores.

Table 9 adds a stricter uncertainty check: instead of only changing named assumption cases, it samples prior distributions over polarization, appointment capture, public pressure, emergency share, justice-pool size, and legislative-output profile components. The table makes close-score caution operational. If two designs' sampled-prior bands overlap, the manuscript treats the comparison as a mechanism contrast rather than an ordinal ranking.

Table 10 states the most important interpretation check directly. If the top cluster changes only

Table 8: Campaign scores and uncertainty bands for selected designs

Scenario	Campaign score	Score 5/50/95	Irregularity 5/50/95	Conflict 5/50/95
18-year terms	0.565	0.515/0.606/0.621	0.059/0.078/0.160	0.338/0.380/0.590
Written emergency reasons	0.562	0.509/0.602/0.618	0.032/0.049/0.123	0.336/0.381/0.591
Emergency merits follow-up	0.560	0.503/0.605/0.620	0.012/0.017/0.053	0.338/0.375/0.596
No emergency merits gap	0.566	0.512/0.609/0.625	0.007/0.011/0.043	0.339/0.373/0.593
Current-like	0.552	0.499/0.590/0.608	0.181/0.239/0.420	0.362/0.408/0.640
Strong recusal enforcement	0.559	0.509/0.599/0.616	0.057/0.077/0.163	0.341/0.381/0.590
Judicial electorate selection	0.554	0.499/0.595/0.612	0.056/0.077/0.163	0.341/0.384/0.595
Constitutional remand	0.566	0.514/0.601/0.617	0.056/0.077/0.164	0.337/0.380/0.591

Table 9: Sampled-prior uncertainty bands for selected designs

Scenario	Score	Rights	Emerg. irregularity	Emerg. down.	Lower ct.	Gov. noncomp.	Conflict	Interpretation
No emergency relief without merits review	0.597/0.615/0.630	0.638/0.648/0.656	0.005/0.009/0.016	0.034/0.058/0.108	0.585/0.606/0.624	0.040/0.065/0.099	0.313/0.358/0.414	front-line cluster
18-year staggered terms + regular appointments	0.589/0.611/0.628	0.630/0.640/0.652	0.051/0.072/0.109	0.043/0.071/0.127	0.575/0.597/0.616	0.037/0.072/0.098	0.317/0.361/0.418	front-line cluster
Automatic merits follow-up for emergency relief	0.591/0.610/0.625	0.634/0.647/0.656	0.009/0.015/0.025	0.041/0.066/0.126	0.583/0.604/0.622	0.044/0.064/0.109	0.314/0.366/0.419	front-line cluster
Mandatory written emergency reasoning	0.586/0.608/0.623	0.626/0.639/0.649	0.029/0.045/0.073	0.049/0.075/0.139	0.574/0.596/0.616	0.047/0.070/0.102	0.320/0.369/0.421	front-line cluster
Constitutional remand before invalidation	0.584/0.607/0.623	0.626/0.640/0.647	0.050/0.074/0.112	0.043/0.071/0.129	0.587/0.609/0.628	0.042/0.066/0.107	0.317/0.368/0.416	front-line cluster
Randomized merits panels with en banc correction	0.584/0.606/0.622	0.631/0.641/0.650	0.052/0.073/0.109	0.044/0.070/0.126	0.575/0.598/0.617	0.042/0.066/0.105	0.319/0.365/0.412	front-line cluster
Independent recusal enforcement with substitutes	0.583/0.606/0.624	0.633/0.641/0.652	0.053/0.074/0.111	0.045/0.071/0.131	0.574/0.598/0.619	0.044/0.069/0.105	0.318/0.365/0.422	front-line cluster
Judicial electorate selection court	0.579/0.601/0.617	0.629/0.644/0.653	0.052/0.074/0.110	0.045/0.073/0.128	0.574/0.597/0.615	0.046/0.071/0.105	0.315/0.368/0.419	front-line cluster
Stylized current U.S.-like supreme court	0.568/0.594/0.616	0.627/0.645/0.658	0.167/0.234/0.324	0.079/0.122/0.196	0.543/0.576/0.602	0.051/0.095/0.150	0.337/0.401/0.462	overlapping uncertainty band

Table 10: Sensitivity drivers and interpretation caveats

Prior	Top-cluster design	Score	Rights	Emerg. irregularity	Emerg. down.	Gov. noncomp.	LC resist.	Caveat
Baseline institutional prior	No emergency relief without merits review	0.610	0.646	0.011	0.055	0.071	0.244	front-line cluster
Baseline institutional prior	60 percent invalidation threshold	0.608	0.633	0.064	0.079	0.070	0.248	rights-protection caveat
High emergency-share prior	18-year staggered terms + regular appointments	0.519	0.620	0.180	0.222	0.225	0.403	compliance caveat
High emergency-share prior	Public-interest litigation filter	0.519	0.630	0.175	0.215	0.218	0.402	compliance caveat
High rights-risk prior	18-year staggered terms + regular appointments	0.527	0.566	0.119	0.135	0.202	0.401	compliance caveat
High rights-risk prior	No emergency relief without merits review	0.527	0.600	0.026	0.098	0.226	0.398	compliance caveat
Weak democratic-mandate prior	60 percent invalidation threshold	0.546	0.604	0.094	0.125	0.180	0.377	compliance caveat
Weak democratic-mandate prior	Constitutional remand before invalidation	0.542	0.616	0.110	0.113	0.176	0.338	compliance caveat
High constitutional-conflict prior	60 percent invalidation threshold	0.512	0.601	0.142	0.194	0.235	0.439	compliance caveat
High constitutional-conflict prior	Constitutional remand before invalidation	0.509	0.614	0.153	0.166	0.235	0.400	compliance caveat
Imported legislative-family prior	No emergency relief without merits review	0.619	0.652	0.009	0.054	0.054	0.229	front-line cluster
Imported legislative-family prior	60 percent invalidation threshold	0.614	0.641	0.060	0.073	0.066	0.234	rights-protection caveat

under high-emergency or high-conflict priors, the paper should describe that result as emergency- or conflict-contingent. If a design remains in the leading cluster but carries a rights, compliance, lower-court-resistance, or emergency-power caveat, the design recommendation should be weakened rather than reported as a clean improvement.

11 Limitations

The model is strongest as a disciplined comparison of institutional mechanisms, not as a predictive empirical model. It does not estimate causal effects from observed constitutional systems. It uses source-backed ranges, comparative institutional coding, and deterministic scenario campaigns to identify where rule bundles create tradeoffs that would otherwise be difficult to see. The correct claim is therefore modest: this is a structured thought experiment with plausibility checks, not a validated model of real court performance.

Several limitations follow directly from that choice. Some emergency results are partly mechanical because the emergency-process-irregularity metric includes opacity, merits follow-through, and emergency opportunism. The paper therefore treats direct reductions in that metric as design implications, not discoveries, unless other outputs move as well. Many parameters are literature-informed or design-prior choices rather than estimated coefficients. The source-range checks are intentionally loose for several constructs, especially emergency denominators, lower-court compliance, comparative access paths, and public-legitimacy proxies. The constitutional-conflict index combines legal, political, implementation, and cross-institutional conflict. Rights-claimant success is not a complete measure of rights protection because some successful claimants may be strategically selected or anti-regulatory, and some rights protection occurs indirectly through structural or administrative cases.

The strongest next validation steps are to automate raw-data refresh for SCDB, Harvard Law Review statistics, shadow-docket data, and recusal data; add one-way sensitivity reports for the highest-leverage emergency, legitimacy, rights, and conflict weight families; calibrate review-path-specific admission denominators against raw certiorari, amparo, constitutional complaint, QPC, abstract-review, and concrete-referral data; add observed counterfactual benchmarks where institutional changes can be compared before and after reform; and benchmark the legislative import contract against raw legislative simulator runs rather than only aggregate campaign CSVs.

A Institutional Parameter Provenance

Appendix Table 12 records how representative metrics should be read. The point is not to create another score table; it is to make the denominator or scale explicit before any empirical, synthetic, or normative inference is drawn.

Appendix Table 13 is a replacement for fine ordinal ranking. It reports alternative reading aids for different constitutional priorities: rights priority, emergency restraint, democratic responsiveness, legal stability, low conflict, and administrative feasibility. These are not new empirical findings. They make visible that any single score embeds a normative weighting choice.

Appendix Table 14 is the audit table for the concern that formal court design might overweight the model. It exposes litigation-pipeline and downstream-enforcement quantities directly: certiorari admission, court-requested response, CVSG request, bar capital, claim strength, vehicle quality, genuine lower-court split rate, lower-court split depth, lower-court resistance, forum shopping, pre-review settlement, strategic plaintiff selection, repeat-player advantage, enforcement capacity, emergency opportunism, recusal incentive pressure, government noncompliance, emergency downstream effect, and precedent durability.

Table 11: Illustrative provenance map for modeled institutional features

Feature family	Model representation	Provenance status
Appointment and tenure	Appointment method, judicial-electorate selector pool, judicial-electorate nominee pool, confirmation threshold, vacancy deadlock risk, term renewability, retirement age, replacement timing	Comparative institutional coding and scenario assumptions
Review access	Access path, petition type, court-requested response, CVSG request, admission score, claimant type, bar capital, claim strength, vehicle quality, lower-court split depth, strategic plaintiff selection, repeat-player advantage, merits transfer, screen-out	Normalized source-register rows and modeled admission rules
Emergency procedure	Application class, relief request, response request, full-court referral, reason giving, status-quo effect, merits follow-through, downstream effect	Shadow-docket source tables plus scenario rules
Decision rule	Voting threshold, panel/en banc routing, quorum failure, recusal consequence, per-remedy thresholds	Court-design coding and institutional assumptions
Post-decision politics	Override attempt, rights carveout, formal response, practical implementation, government noncompliance, open defiance	Comparative public-law coding and strategic actor assumptions
Outputs	Stability, rights protection, claimant success, legitimacy, conflict, responsiveness, precedent durability, lower-court compliance	Simulated outcomes with calibration guardrails

B Model Mechanics

This appendix records the main state machine and scoring formulas so the manuscript does not treat the simulator as a black box. Each run draws a filed universe from a shared world seed. For each scenario, the same filed matters are reviewed by a court process with a scenario-specific design and a scenario-specific random stream. Let i index filed matters and s index institutional scenarios.

Admission begins with a path-specific petition probability. In the compact notation below, c is legal conflict, a is public attention, g is SG signal, d is split depth, k is bar capital, m is claim strength, r is repeat-player advantage, p is strategic-plaintiff selection, f is forum-shopping pressure,

Table 12: Metric semantics and manuscript use

Family	Metric	Use	Compatibility	Manuscript interpretation
certiorari	paid cert petition share	loose calibration	near pathway match	conditional paid/IFP split is closer than all-case paidPetitionRate, but the simulator still excludes many non-review docketed matters
certiorari	certiorari admission rate	proxy context	denominator mismatch	the source includes all docketed matters; the simulator denominator is already a constitutional-review cert-path subset
emergency	emergency grant conditional rate	loose calibration	conditional near match	conditional rate is closer than all-case emergencyGrantRate, but the simulated emergency universe is not limited to noncapital applications
complaint referral	admission rate	proxy context	denominator mismatch	this is a strong complaint-filter source but not a whole-universe admission target
lower court compliance	lower court compliance	proxy context	mechanism proxy	the source anchors direction and magnitude of alignment pressure, not a compliance-level target
lower court compliance	government noncompliance rate	loose calibration	mechanism proxy	closest available implementation anchor, but still a study-sample proxy rather than a court-wide noncompliance rate
override remand	override rate	design prior	design prior	duration informs institutional design, not observed behavioral override success
headline	directional score	reading aid	not empirical target	used only for ordering and clustering; close score differences are not substantive rankings
headline	rights claimant success	synthetic output	not empirical target	direct model output, not blended into empirical validation
headline	emergency process irregularity	synthetic output	not empirical target	constructed emergency-process index, not an adjudicative finding of legal abuse
headline	process legitimacy proxy	synthetic output	not empirical target	modeled process-legitimacy proxy, not public-opinion measurement
headline	constitutional conflict	synthetic output	not empirical target	mechanism summary metric; individual channels remain reported separately

Table 13: Multi-objective reading aids under different normative priorities

Scenario	Rights priority	Emergency restraint	Democratic response	Legal stability	Low conflict	Admin feasibility
Current-like	0.475	0.437	0.591	0.609	0.700	0.668
18-year terms	0.455	0.627	0.629	0.620	0.730	0.691
Written reasons	0.450	0.620	0.629	0.625	0.726	0.689
Auto merits	0.506	0.685	0.640	0.598	0.734	0.691
No merits gap	0.495	0.690	0.646	0.605	0.738	0.696
Recusal enforce.	0.457	0.627	0.635	0.619	0.733	0.679
Judicial electorate	0.456	0.628	0.636	0.620	0.734	0.674
Random panels	0.452	0.627	0.639	0.622	0.731	0.661
Integrity package	0.500	0.685	0.647	0.603	0.738	0.684
Constitutional remand	0.436	0.629	0.641	0.651	0.742	0.672
Public-interest filter	0.465	0.629	0.634	0.624	0.734	0.681

Table 14: Litigation-pipeline and downstream-enforcement diagnostics

Scenario	Cert admit	CFR	CVSG	Bar	Claim	Vehicle	Genuine split	Split	Resistance	Forum	Settlement	Plaintiff sel.	Repeat player	Enforcement	Emerg. opp.	Recusal press.	Gov. noncomp.	Emerg. down.	Prec. dur.
Current-like	0.195	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.332	0.549	0.005	0.549	0.384	0.552	0.570	0.491	0.181	0.186	0.599
18-year terms	0.190	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.506	0.007	0.549	0.384	0.561	0.417	0.497	0.147	0.112	0.633
Written emergency reasons	0.196	0.119	0.001	0.570	0.575	0.681	0.535	0.589	0.322	0.506	0.007	0.549	0.384	0.561	0.391	0.497	0.149	0.128	0.649
Automatic merits follow-up	0.194	0.119	0.001	0.570	0.575	0.681	0.535	0.589	0.317	0.507	0.007	0.549	0.384	0.567	0.369	0.497	0.148	0.089	0.588
No emergency merits gap	0.197	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.317	0.507	0.007	0.549	0.384	0.567	0.349	0.509	0.146	0.088	0.605
Strong recusal enforcement	0.194	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.506	0.007	0.549	0.384	0.561	0.418	0.513	0.148	0.113	0.632
Judicial electorate selection	0.199	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.506	0.007	0.549	0.384	0.561	0.417	0.500	0.149	0.114	0.634
Random merits panels	0.193	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.476	0.007	0.549	0.384	0.561	0.418	0.604	0.146	0.114	0.639
Emergency integrity package	0.196	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.317	0.507	0.006	0.549	0.384	0.567	0.369	0.513	0.146	0.089	0.598
Jurisdiction-stripping constraints	0.195	0.117	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.456	0.007	0.549	0.384	0.591	0.417	0.497	0.161	0.113	0.633
Legislative override window	0.194	0.119	0.001	0.570	0.575	0.681	0.535	0.589	0.322	0.507	0.007	0.549	0.384	0.563	0.417	0.497	0.156	0.113	0.633
Constitutional remand	0.201	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.286	0.465	0.007	0.549	0.384	0.580	0.418	0.501	0.145	0.113	0.698
Public-interest filter	0.200	0.117	0.001	0.570	0.575	0.681	0.535	0.589	0.321	0.421	0.007	0.549	0.384	0.562	0.417	0.501	0.147	0.112	0.643
Remand + override window	0.197	0.119	0.001	0.570	0.575	0.681	0.535	0.589	0.286	0.465	0.006	0.549	0.384	0.580	0.391	0.501	0.146	0.128	0.711
Panel + jurisdiction safeguards	0.197	0.118	0.001	0.570	0.575	0.681	0.535	0.589	0.322	0.425	0.006	0.549	0.384	0.589	0.402	0.631	0.156	0.131	0.669

q is settlement pressure, and v is vehicle-defect risk:

$$P_{is} = \mathrm{clamp}_{[0,1]}(b_{\mathrm{petition}} + .08c_i + .08a_i + .06g_i + .08d_i + .05k_i + .05m_i + .06r_i + .08p_i + .04f_i - .05q_i - .22v_i).$$

Conditional on a petition or referral, the admission score combines certiorari pressure, lower-court conflict, lower-court error, lower-court split depth, genuine split status, SG signal, court-requested response, CVSG request, amicus intensity, split maturity, relist signal, claimant type, bar capital, claim strength, vehicle quality, specialist counsel, repeat-player advantage, strategic plaintiff selection, forum-shopping pressure, settlement pressure, conditional reversal probability, access-path adjustment, institutional gatekeeping, vehicle defects, and vacancy deadlock. Merits transfer is then path-specific: emergency applications transfer through the application's merits-follow-through probability, complaints and amparo transfer at a lower rate, and ordinary referrals or cert-like paths transfer at a higher baseline reduced by settlement pressure.

After admission, the court process chooses strategic-actor response, emergency status, pre-review settlement, initial merits review, panel or en banc routing, recusals, quorum, votes, emergency order, invalidation, auxiliary-review outcome, override attempt, and post-decision implementation response. The direct metrics are case averages over the filed universe. For example, rights-claimant success is one when a high-rights-burden claimant receives invalidation or emergency relief and partial when settlement or constitutional remand preserves a claimant's route without immediate invalidation; the modeled public-legitimacy proxy is legitimacy minus penalties for emergency legitimacy risk, partisan alignment, appointment-pressure campaigns, and court-curbing response; the constitutional-conflict index rises with conflict potential, executive defiance, lower-court conflict, lower-court resistance, accumulated state conflict, government noncompliance, emergency downstream effect, invalidating high-mandate laws, emergency relief, cross-court disagreement, override attempts, and repeated override loops.

The directional score is intentionally secondary:

$$\begin{aligned}\mathrm{DirectionalScore}_s = \mathrm{mean}(&\mathrm{StabilityRights}_s, \mathrm{LegitimacyControl}_s, \mathrm{ClaimantSuccess}_s,\\ &\mathrm{PrecedentDurability}_s, \mathrm{LowerCourtCompliance}_s, \mathrm{EliteAcceptance}_s,\\ &1 - \mathrm{AdministrativeCost}_s).\end{aligned}$$

The stability-rights component averages legal stability, precedent durability, lower-court compliance, enforcement capacity, lower-court-resistance restraint, rights protection, reversal restraint, and conflict restraint. The legitimacy-control component averages legitimacy, democratic responsiveness, independence-accountability balance, partisan-alignment restraint, emergency-irregularity restraint, emergency-legitimacy-risk restraint, emergency-downstream restraint, government-noncompliance

Table 15: Representative model weights and decision rules

Component	Representative rule	Interpretation
Admission	0.18 + .22 cert pressure +.12 lower-court conflict +.10 lower-court error +.13 split depth +.06 genuine split +.15 SG signal +.10 amicus signal +.08 split maturity +.07 relist signal +.06 bar capital +.10 claim strength +.09 vehicle quality +.05 specialist counsel +.05 repeat-player signal +.05 strategic-plaintiff signal +.02 forum-shopping signal +.08 conditional reversal −.06 settlement pressure −.16 vehicle defect, plus path adjustment, CFR, and CVSG increments	Admission is an institutional screen, not a fixed docket quota.
Voting	Legal concern receives weight .42; rights concern uses rights burden, justice rights sensitivity, and weight .44; claim strength, vehicle quality, and bar capital can modestly increase merits credibility; democratic mandate, legislative quality, institutionalism, emergency deference, repeat-player pressure, forum-shopping signal, precedent stability, and accountability pull reduce or increase invalidation propensity.	Merits votes mix legal, rights, strategic, institutional, and emergency-deference considerations.
Emergency irregularity	Open emergency rule base .32, no-merits penalty .22, shadow stay .24, written reasons −.10, automatic merits follow-up −.12, merits acceleration −.07, partisan salience .12, ambiguity .08, emergency opportunism .10.	The legacy `shadowDocketAbuse` field is treated as an emergency-process-irregularity index driven by opacity, opportunistic filing, and merits-like relief without ordinary merits process.
Legitimacy proxy	Base .48, public attention .16, legislative quality .10, merits reason-giving .18, recusal discipline up to .16, council warning .06, emergency irregularity −.24, partisan alignment −.20.	The modeled public-legitimacy proxy rewards visible process and penalizes opaque or partisan-looking intervention; it is not direct public-opinion measurement.
Compliance	Lower-court compliance rises with enforcement capacity and merits clarity, and falls with lower-court conflict, error risk, split depth, ideological drift, lower-court resistance, state conflict, invalidation shock, and government noncompliance.	Enforcement and implementation are modeled separately from formal legal validity.
Conflict	Conflict potential .38, executive defiance .18, lower-court conflict .08, lower-court resistance .10, forum shopping .04, conflict load .12, legislative defiance .10, government noncompliance, emergency downstream effect, invalidating high-mandate laws .14, shadow relief .12, cross-court disagreement .16, override attempt .06.	Constitutional conflict is cumulative and rises when legal disagreement becomes interbranch disagreement.
Override	Attempt pressure combines override pressure .34, democratic mandate .30, public attention .18, conflict load .12, legislative defiance .12, override adaptation .10, and rights-burden penalty −.24.	Accountability devices affect implementation only after invalidation and can generate rights carveouts or repeated override loops.

restraint, recusal-pressure restraint, forum-shopping restraint, emergency-opportunism restraint, and strategic-pressure restraint. Because those choices are normative, the paper reports the direct outputs separately and uses the directional score only as a reading aid. Score differences below .010 are treated as indistinguishable in the manuscript's substantive interpretation.

Data Availability Statement

The simulator, normalized calibration tables, frozen legislative-output fixtures, deterministic seeds, scenario definitions, generated reports, figure-generation scripts, standalone figure exports, source-audit file (`paper/source-audit.csv`), source-provenance manifests, and report manifests are maintained as a reproducible project. Public repository identifiers are withheld in this anonymous review manuscript. The intended replication package includes source code, build instructions, normalized input data, generated campaign artifacts, generated figure fragments, standalone figure files, manifest hashes, and documentation distinguishing observed inputs, coding choices, model assumptions, and simulated outputs. For final publication, the package should be deposited in the Journal of Law and Courts Dataverse or another repository linked from the JLC Dataverse rather than being available only on request.

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